Geometry Review: Unit 4 — Similarity

Station 1 of 4: Ratios and Proportions

Name: _____ Date: _____

Solve each proportion or ratio problem. Show your cross products.

1. Solve: $\frac{x}{4} = \frac{15}{20}$

2. Solve: $\frac{6}{x} = \frac{9}{12}$

3. Solve: $\frac{x+3}{8} = \frac{6}{4}$

4. Solve: $\frac{3}{x-1} = \frac{6}{10}$

5. Write the ratio of 18 to 24 as a simplified fraction.

6. The angles of a triangle are in the ratio $2 : 3 : 4$. Find each angle.

7. Two numbers are in the ratio $5 : 7$ and their sum is 84. Find the numbers.

8. Is $\frac{6}{9} = \frac{10}{15}$ a true proportion? Explain.

9. A recipe uses 3 cups of flour for 24 cookies. How many cups are needed for 40 cookies?

10. On a map, 1 inch represents 25 miles. How many miles do 3.5 inches represent?

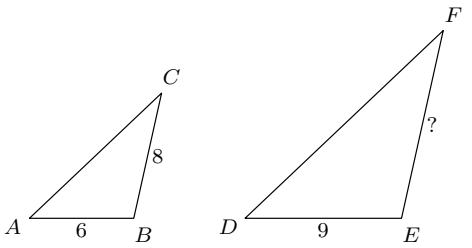
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Station 2 of 4: Similar Polygons and Scale Factor

Name: _____ Date: _____

Use scale factors and proportions. Remember: area ratios are the square of the scale factor. Diagrams are not drawn to scale.

1. $\triangle ABC \sim \triangle DEF$ with $AB = 6$, $DE = 9$, and $BC = 8$. Find EF.



2. Two similar figures have corresponding sides 4 and 10. What is the scale factor from the smaller to the larger figure?

3. A 4×6 rectangle is similar to a $10 \times x$ rectangle. Find x.

4. $\triangle ABC \sim \triangle XYZ$, $AB = 10$, $XY = 15$, and the perimeter of $\triangle ABC$ is 24. Find the perimeter of $\triangle XYZ$.

5. $\triangle ABC \sim \triangle DEF$ and $m\angle A = 50°$. Find $m\angle D$.

6. Two similar quadrilaterals have a scale factor of $2 : 5$. The longest side of the smaller one is 14. Find the longest side of the larger one.

7. Are a 3×5 rectangle and a 6×9 rectangle similar? Explain.

8. Are triangles with sides $3, 4, 5$ and $9, 12, 15$ similar? If so, give the scale factor.

9. Two similar triangles have a scale factor of $3 : 5$. What is the ratio of their areas?

10. A triangle has area 20. A similar triangle has sides 3 times as long. Find its area.

Geometry Review: Unit 4 — Similarity

Station 3 of 4: Proving Triangles Similar: AA, SAS, SSS

Name: _____ Date: _____

Decide whether the triangles are similar. Name the theorem (AA, SAS∼, SSS∼) or write "not similar." Diagrams are not drawn to scale.

1. $\triangle 1$ has angles $50°$ and $60°$. $\triangle 2$ has angles $60°$ and $70°$. Are they similar?

2. Triangle sides $4, 6, 8$ and $6, 9, 12$. Similar? If so, which theorem and what scale factor?

3. In $\triangle ABC$, $AB = 6$, $AC = 9$, $m\angle A = 40°$. In $\triangle DEF$, $DE = 4$, $DF = 6$, $m\angle D = 40°$. Similar?

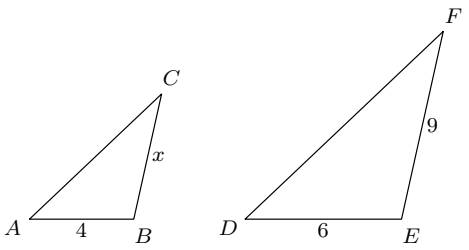
4. Triangle sides $5, 7, 9$ and $10, 14, 20$. Similar?

5. Two right triangles each have an acute angle of $35°$. Are they similar?

6. Given $\angle A \cong \angle D$ and $\angle B \cong \angle E$, which theorem proves $\triangle ABC \sim \triangle DEF$?

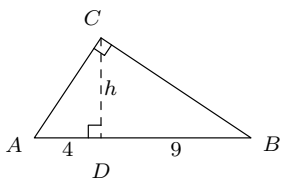
7. Given $\angle A \cong \angle D$ and $\dfrac{AB}{DE} = \dfrac{3}{2}$, what additional ratio is needed to prove similarity by SAS∼?

8. $\triangle ABC \sim \triangle DEF$ with $AB = 4$, $DE = 6$, $BC = x$, and $EF = 9$. Find x.



9. Triangles with sides $6, 8, 10$ and $9, 12, 15$: similar? Scale factor?

10. The altitude to the hypotenuse of a right triangle splits the hypotenuse into segments of 4 and 9. Find the altitude.



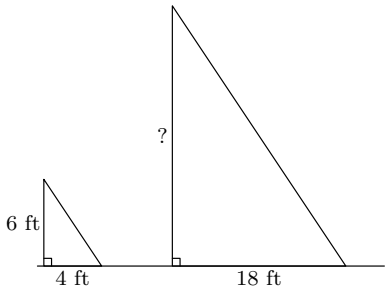
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Station 4 of 4: Applications of Similarity

Name: _____ Date: _____

Draw a sketch if it helps. Set up a proportion for each problem. Diagrams are not drawn to scale.

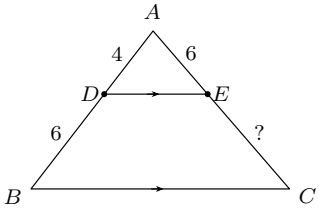
1. A 6-ft person casts a 4-ft shadow. At the same time a tree casts an 18-ft shadow. How tall is the tree?



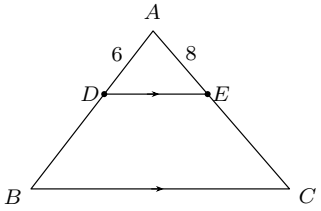
2. A 5-ft stick casts a 3-ft shadow. At the same time a flagpole casts a 24-ft shadow. How tall is the flagpole?

3. A 2-m stick casts a 1.5-m shadow. At the same time a building casts a 30-m shadow. How tall is the building?

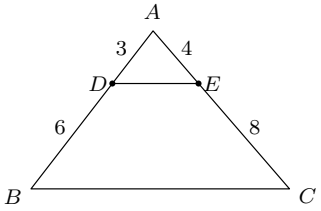
4. In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$ with D on $\overline{AB}$ and E on $\overline{AC}$. If $AD = 4$, $DB = 6$, and $AE = 6$, find EC.



5. In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. If $AD = 6$, $AB = 15$, and $AE = 8$, find AC.



6. In $\triangle ABC$, D is on $\overline{AB}$ and E is on $\overline{AC}$ with $AD = 3$, $DB = 6$, $AE = 4$, $EC = 8$. Is $\overline{DE} \parallel \overline{BC}$? Explain.



7. The geometric mean of 8 and 18 is x, where $\dfrac{8}{x} = \dfrac{x}{18}$. Find x.

8. A model car is built at a scale of 1 : 50. The model is 0.6 m tall. How tall is the actual object?

9. Two similar triangles have a ratio of perimeters of 7 : 4. The smaller has perimeter 28. Find the perimeter of the larger.

10. Two similar triangles have areas 25 and 49. The shorter side of the smaller triangle is 10. Find the corresponding side of the larger triangle.

Geometry Review: Unit 4 — Similarity

Answer Key

Station 1: Ratios and Proportions

1. $x = 3$
2. $x = 8$
3. $x = 9$
4. $x = 6$
5. $\frac{3}{4}$
6. $40°,\ 60°,\ 80°$
7. 35 and 49
8. Yes; $6 \cdot 15 = 9 \cdot 10 = 90$
9. 5 cups
10. 87.5 miles

Station 2: Similar Polygons and Scale Factor

1. $EF = 12$
2. $\frac{5}{2}$
3. $x = 15$
4. 36
5. 50°
6. 35
7. No; $\frac{3}{6} \neq \frac{5}{9}$
8. Yes; scale factor 3
9. $9 : 25$
10. 180 $\quad 20 \cdot 3^2$

Station 3: Proving Triangles Similar: AA, SAS, SSS

1. Yes (AA); the third angles are 70° and 50°
2. Yes; SSS∼, scale factor $\frac{3}{2}$
3. Yes; SAS∼ $(\frac{6}{4} = \frac{9}{6} = \frac{3}{2})$
4. No; $\frac{5}{10} = \frac{7}{14} = \frac{1}{2}$ but $\frac{9}{20} \neq \frac{1}{2}$
5. Yes (AA)
6. AA
7. $\frac{AC}{DF} = \frac{3}{2}$
8. $x = 6$
9. Yes; SSS∼, $\frac{3}{2}$
10. 6 $\quad h^2 = 4 \cdot 9$

Station 4: Applications of Similarity

1. 27 ft
2. 40 ft
3. 40 m
4. $EC = 9$
5. $AC = 20$
6. Yes; $\frac{3}{6} = \frac{4}{8} = \frac{1}{2}$
7. $x = 12$
8. 30 m
9. 49
10. 14 $\quad$ *side ratio* $= \sqrt{25} : \sqrt{49} = 5 : 7$